

5T2: Sinusoidal model

(2 of 3)

Xavier Serra

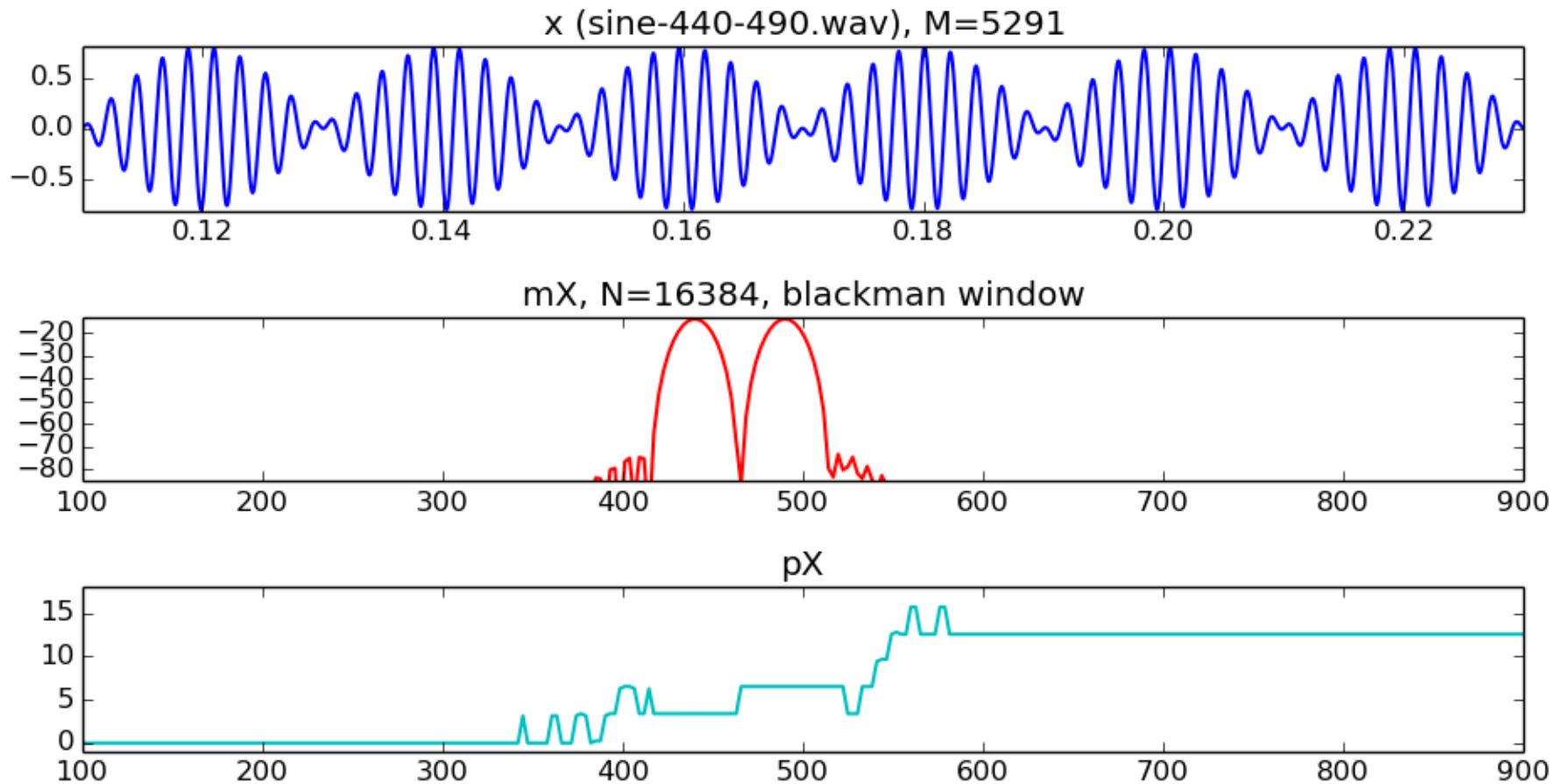
Universitat Pompeu Fabra, Barcelona

Index

- Sinusoidal model
- Peak detection
- Time-varying sinewaves in spectrogram

Sinusoidal model

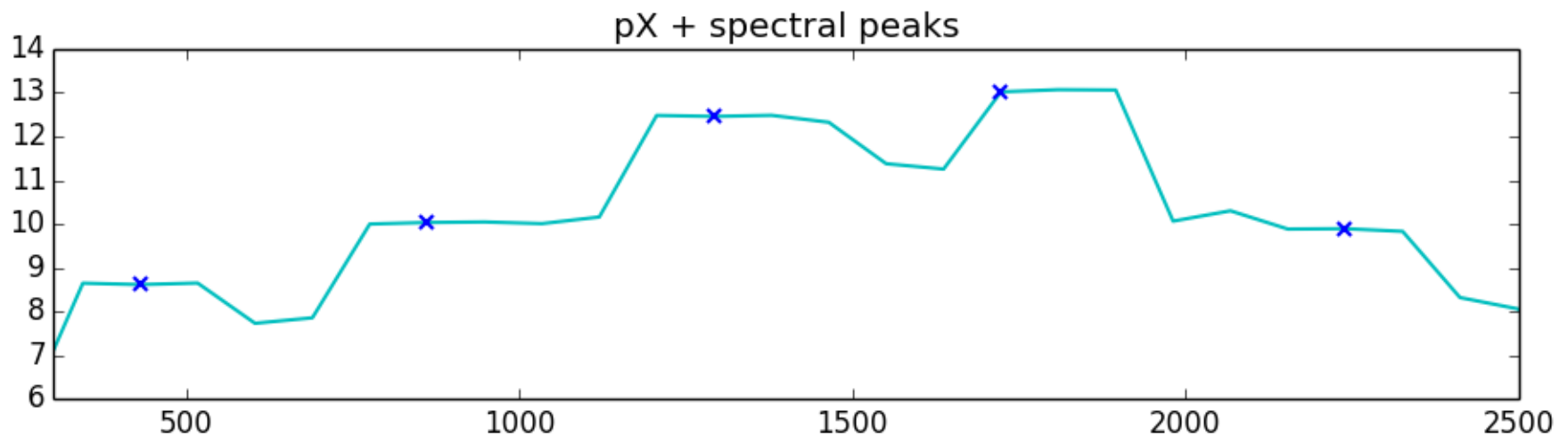
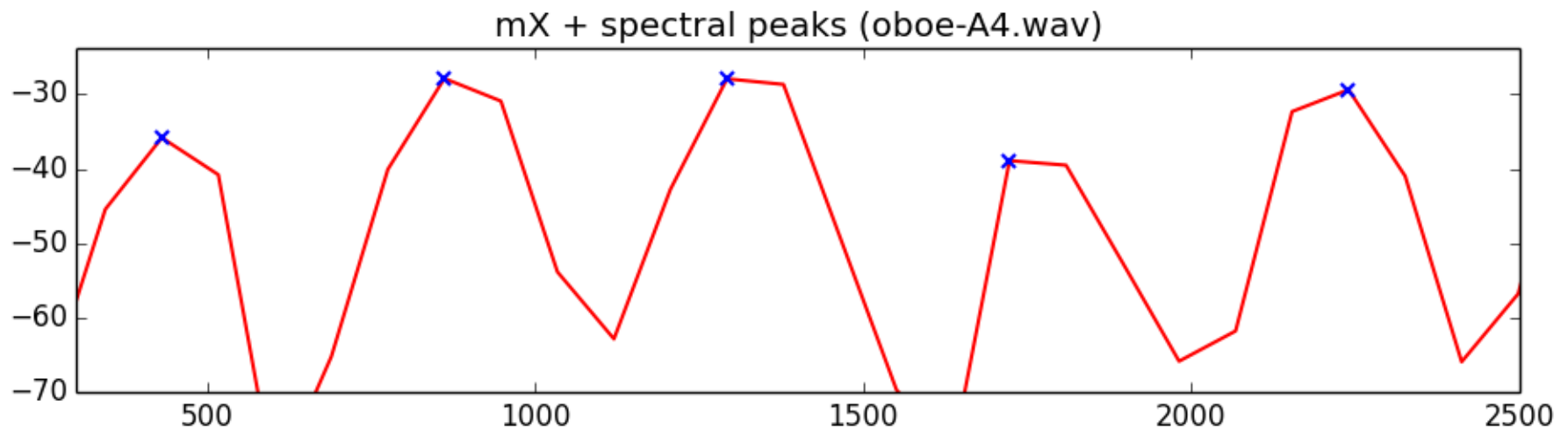
$$y[n] = \sum_{r=1}^R A_r[n] \cos(2\pi f_r[n]n)$$



Peak detection

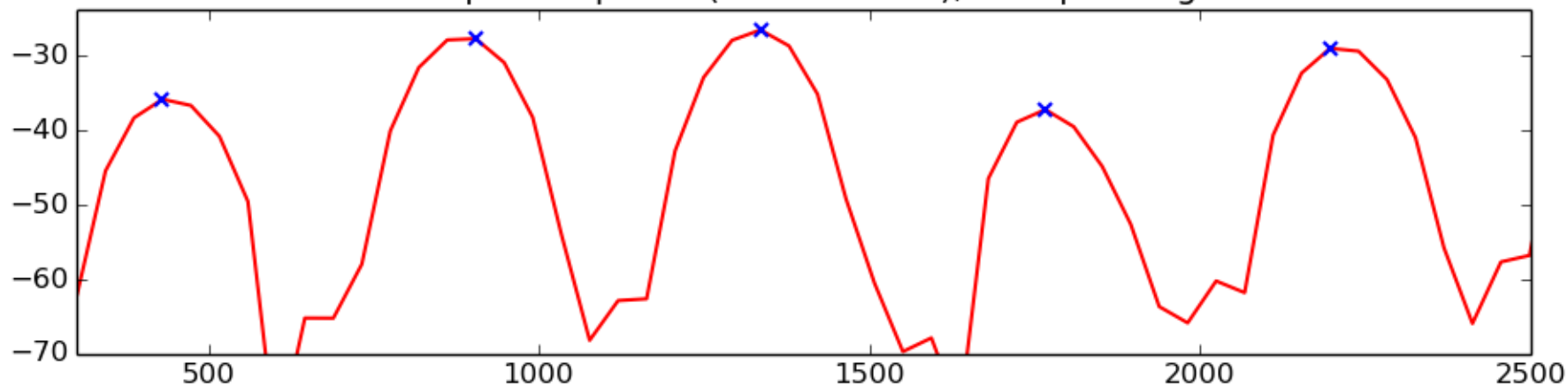
$$p_r = mX[k_0] \text{ when } mX[k_0 - 1] < mX[k_0] > mX[k_0 + 1]$$

mX = magnitude spectrum ; r = peak number ; k_0 = peak location

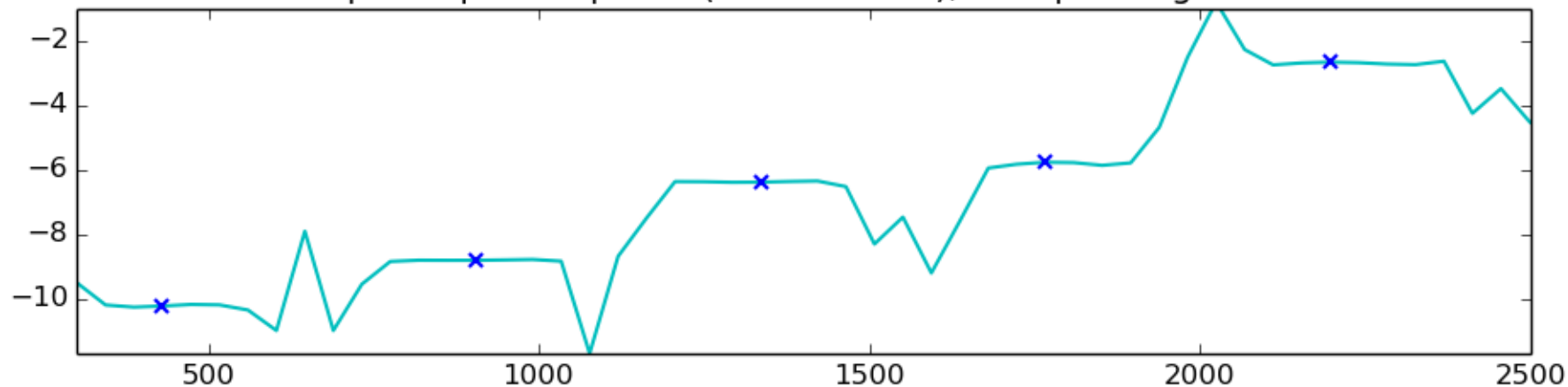


Peak detection with zero-padding

mX + spectral peaks (oboe-A4.wav), zero padding = 2



pX + spectral peaks (oboe-A4.wav), zero padding = 2



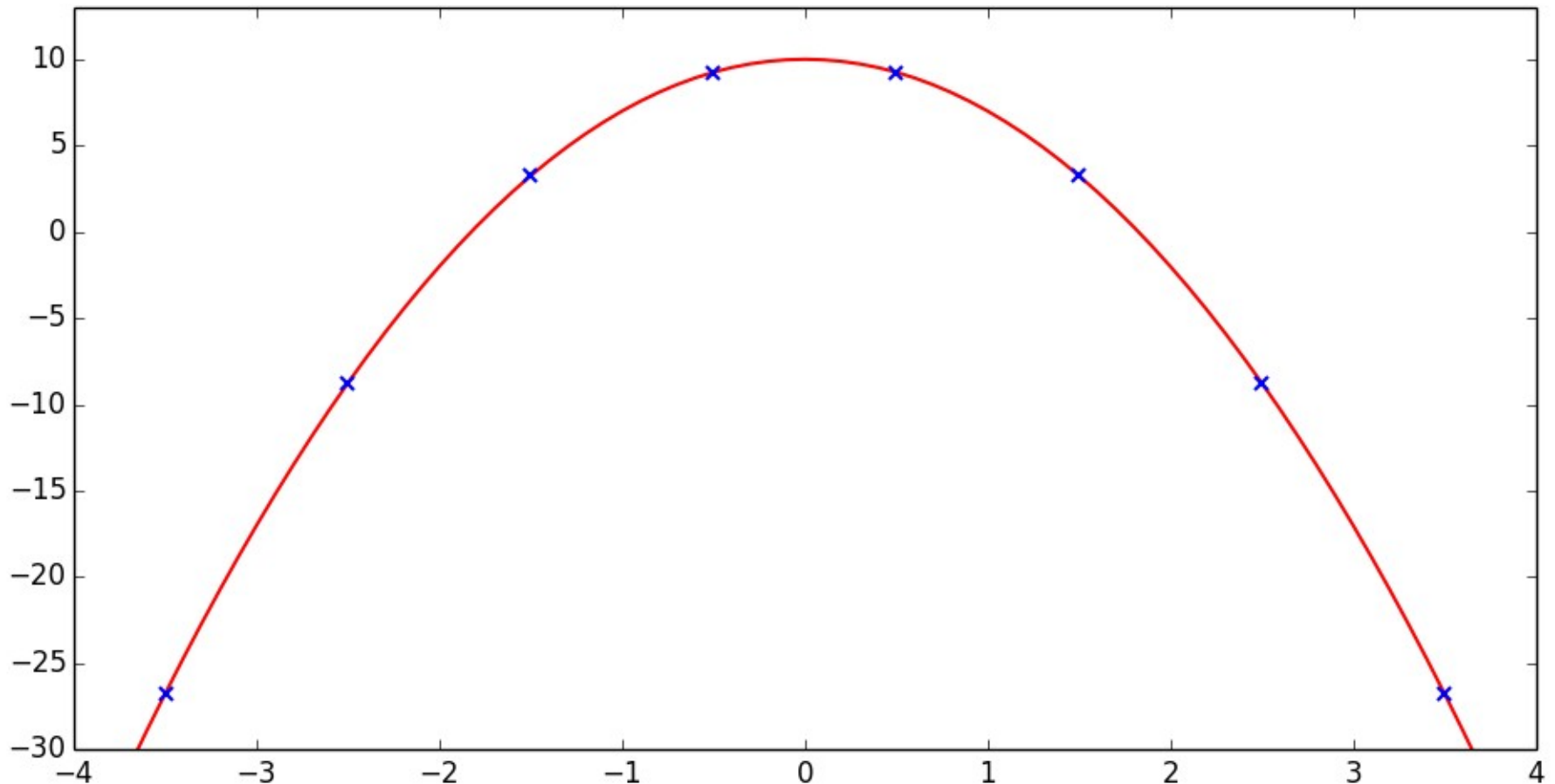
Parabola

$$x[n] = a(n - p)^2 + b$$

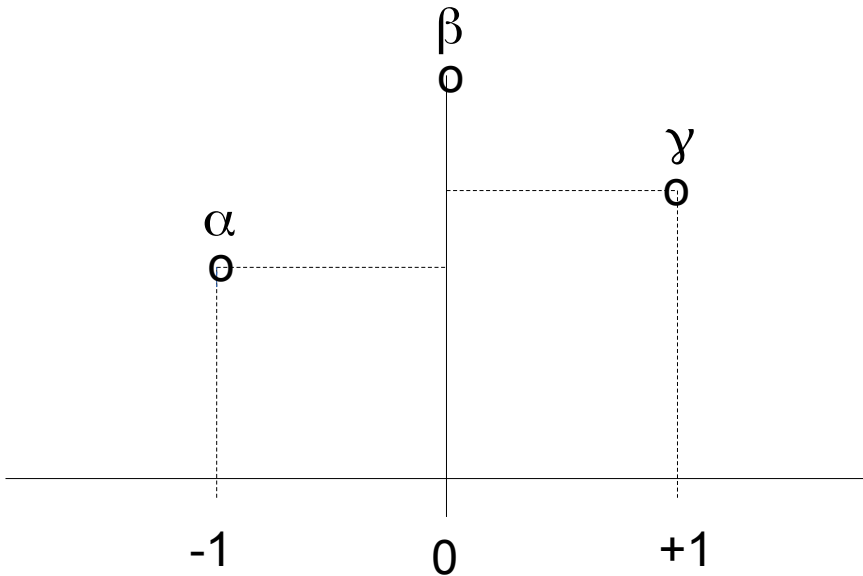
p : center of parabola

a : concavity measure

b : offset



Peak interpolation



$$x[-1] = \alpha = mX[k_{\beta}-1]$$

$$x[0] = \beta = mX[k_{\beta}]$$

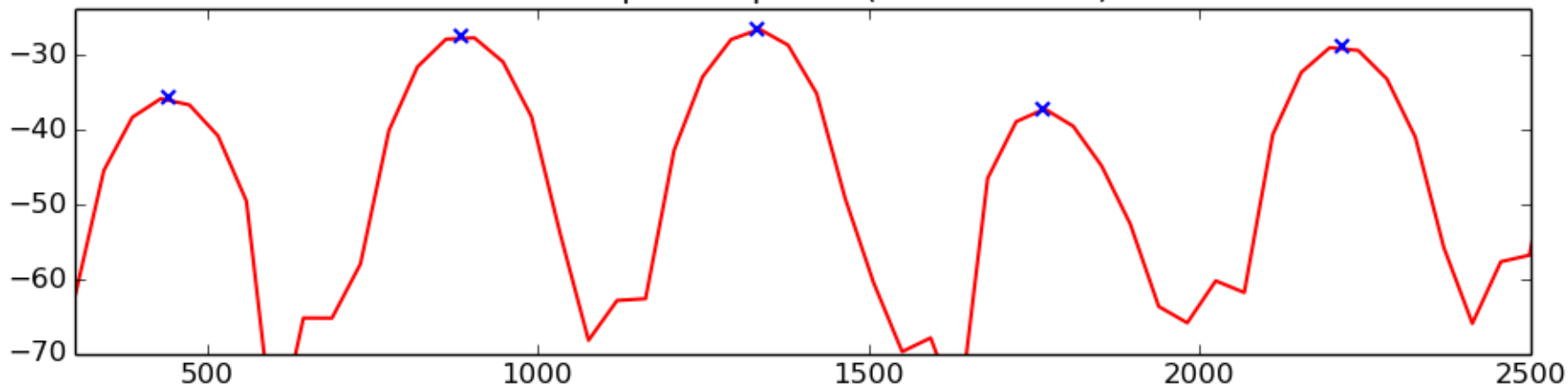
$$x[1] = \gamma = mX[k_{\beta}+1]$$

center of the parabola: $\hat{k}_p = k_p + \frac{1}{2} \frac{\alpha - \gamma}{(\alpha - 2\beta + \gamma)}$

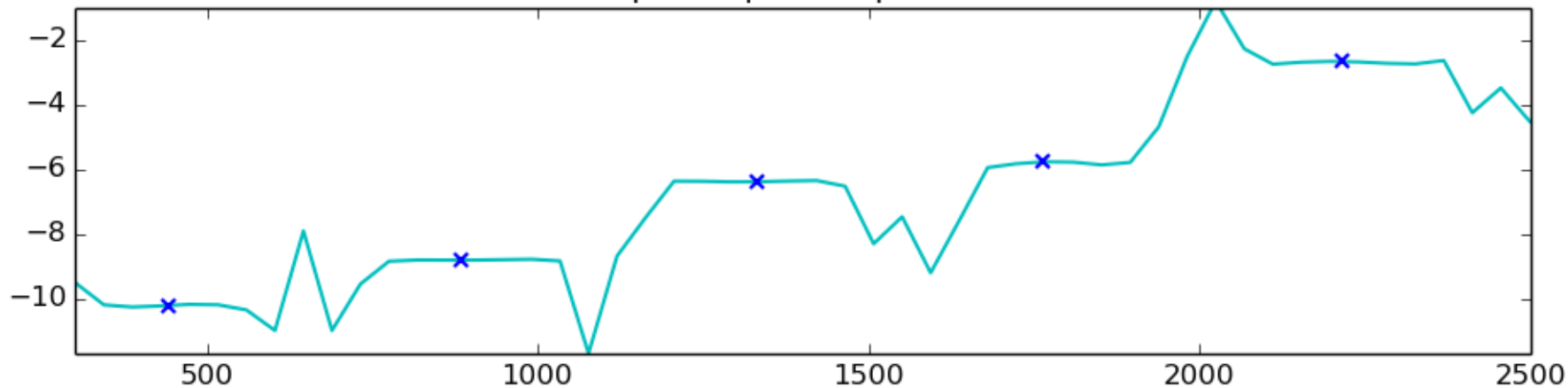
amplitude: $\hat{a} = \beta - \frac{1}{4} (\hat{k}_p - k_p)(\alpha - \gamma)$

Peak detection with interpolation

mX + spectral peaks (oboe-A4.wav)



pX + spectral peaks



Sinusoidal parameters from peaks

$$\hat{k}_p = k_p + \frac{0.5 * (|X[k_p - 1]| - |X[k_p + 1]|)}{|X[k_p - 1]| - 2 * |X[k_p]| + |X[k_p + 1]|}$$

$$f_p = \frac{f_s * \hat{k}_p}{N}$$

$$A_p = |X[k_p]| - 0.25 * (|X[k_p - 1]| - |X[k_p + 1]|) * (\hat{k}_p - k_p)$$

$$ph_p = \angle X[\hat{k}_p]$$

Sinewaves in spectrogram

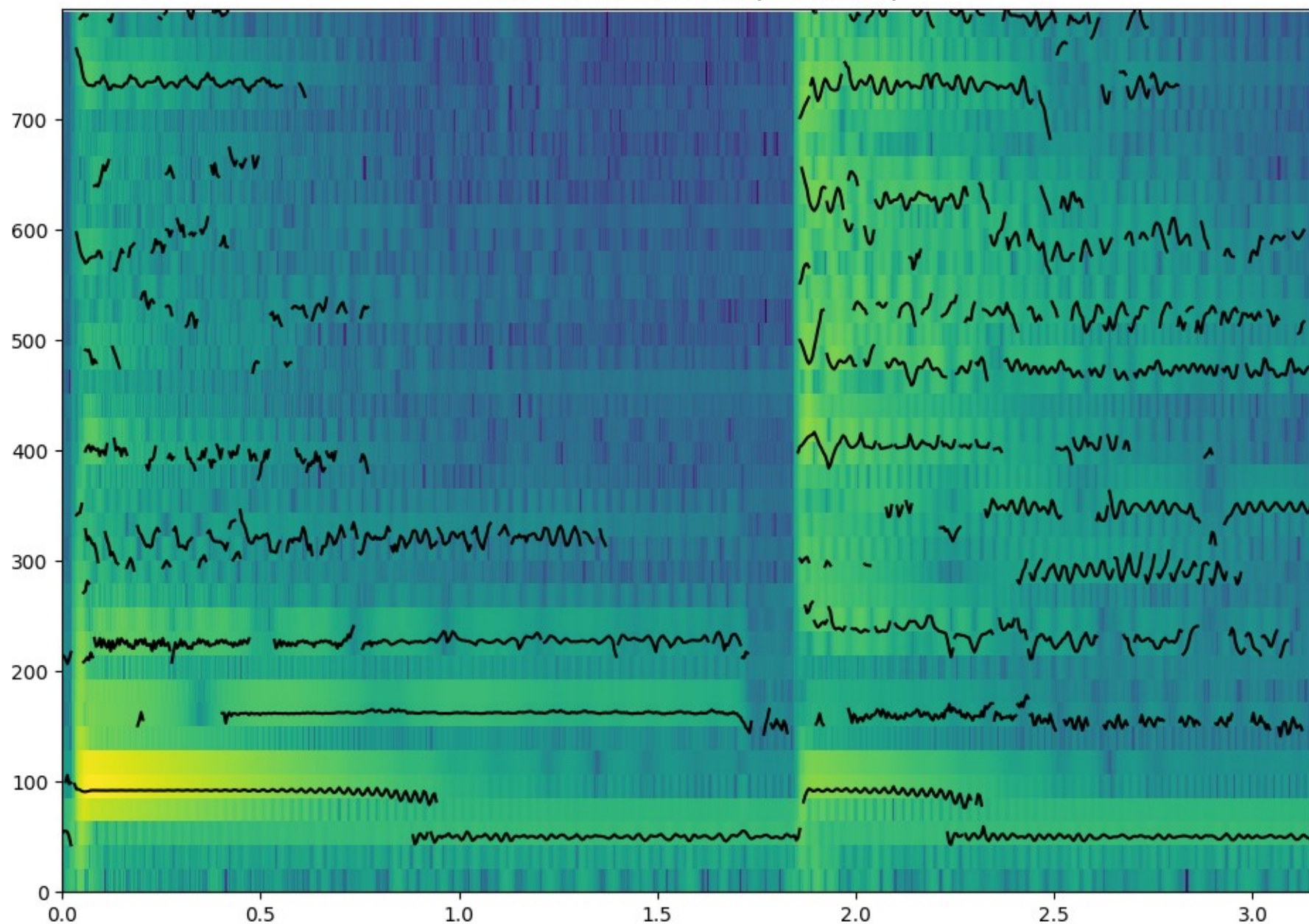
- Sinusoid → **stable** peak track in spectrogram
- Stability identified by
 - frequency and amplitude in successive frames
 - phase derivative in time/freq

Condition for a peak f_p of frame l to be part of a track t :

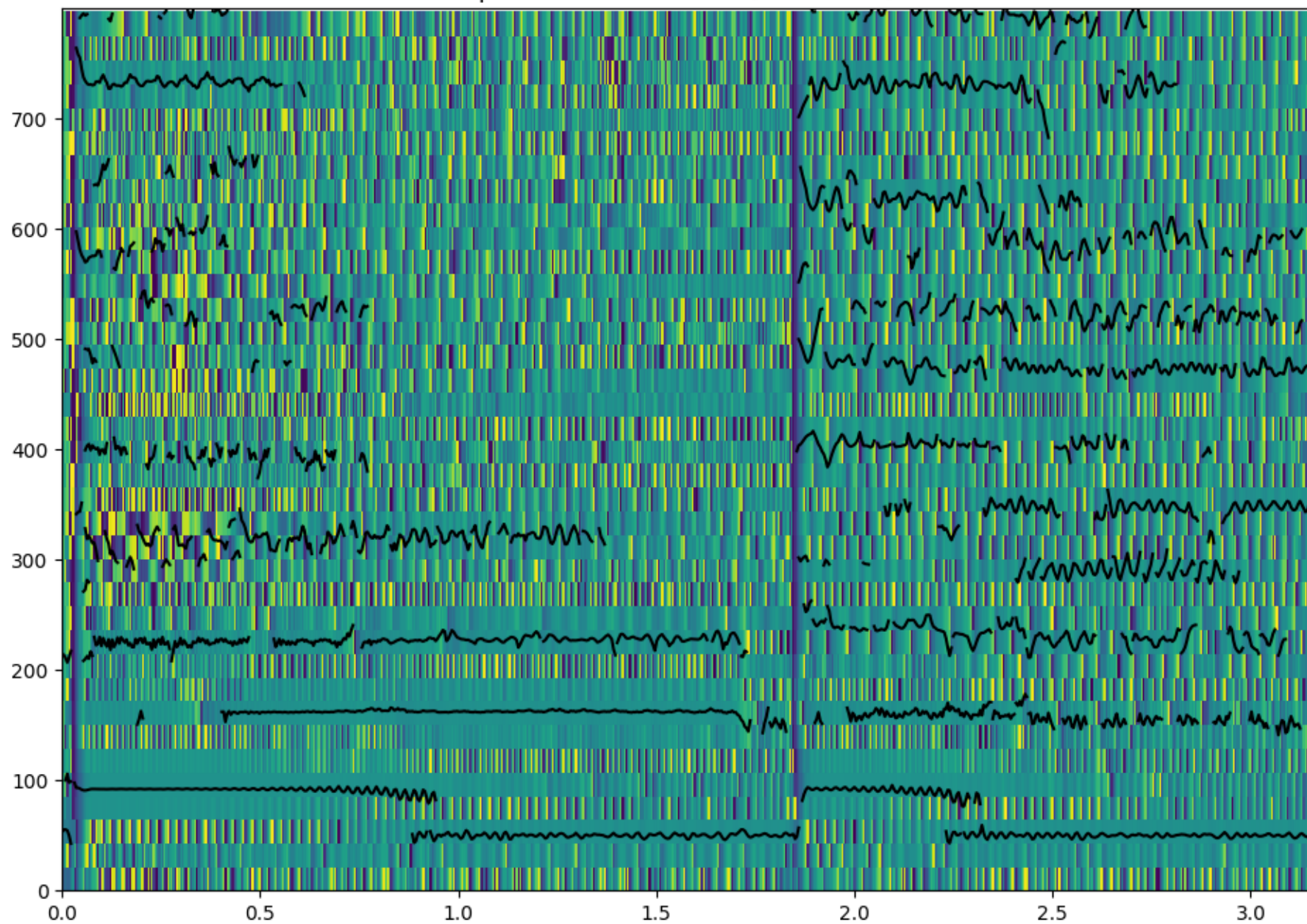
$$f_p[l] = f_t[l] \text{ if } (|f_p[l] - f_t[l-1]| < \textit{threshold})$$

and if exists $f_t[l-2], f_t[l-3], \dots, f_t[l-L]$

mX + sinusoidal tracks (bendir.wav)



pX + sinusoidal tracks (bendir.wav)



References and credits

- More information in:
http://en.wikipedia.org/wiki/Sinusoidal_model
- Reference on sinusoidal modeling by Julius O. Smith:
https://ccrma.stanford.edu/~jos/sasp/Spectrum_Analysis_Sinusoids.html
- Sounds from:
<http://www.freesound.org/people/xserra/packs/13038/>
- Slides released under CC Attribution-Noncommercial-Share Alike license and code under the Affero GPL license; available from <https://github.com/MTG/sms-tools>

5T2: Sinusoidal model

(2 of 3)

Xavier Serra

Universitat Pompeu Fabra, Barcelona